

Description: The objective of this assignment is to use an ultrasonic sensor, combined with a servo motor, to detect obstacles in the way of the robots path. If there is nothing in something in the way while going straight, it will turn right. If there is an object detected on the left, the robot will turn right.

Implementation: This project required multiple jumper cables, motor driver, FRDM board, 2 DC motors, 1 servo motor, and 1 ultrasonic sensor.

Pseudocod:

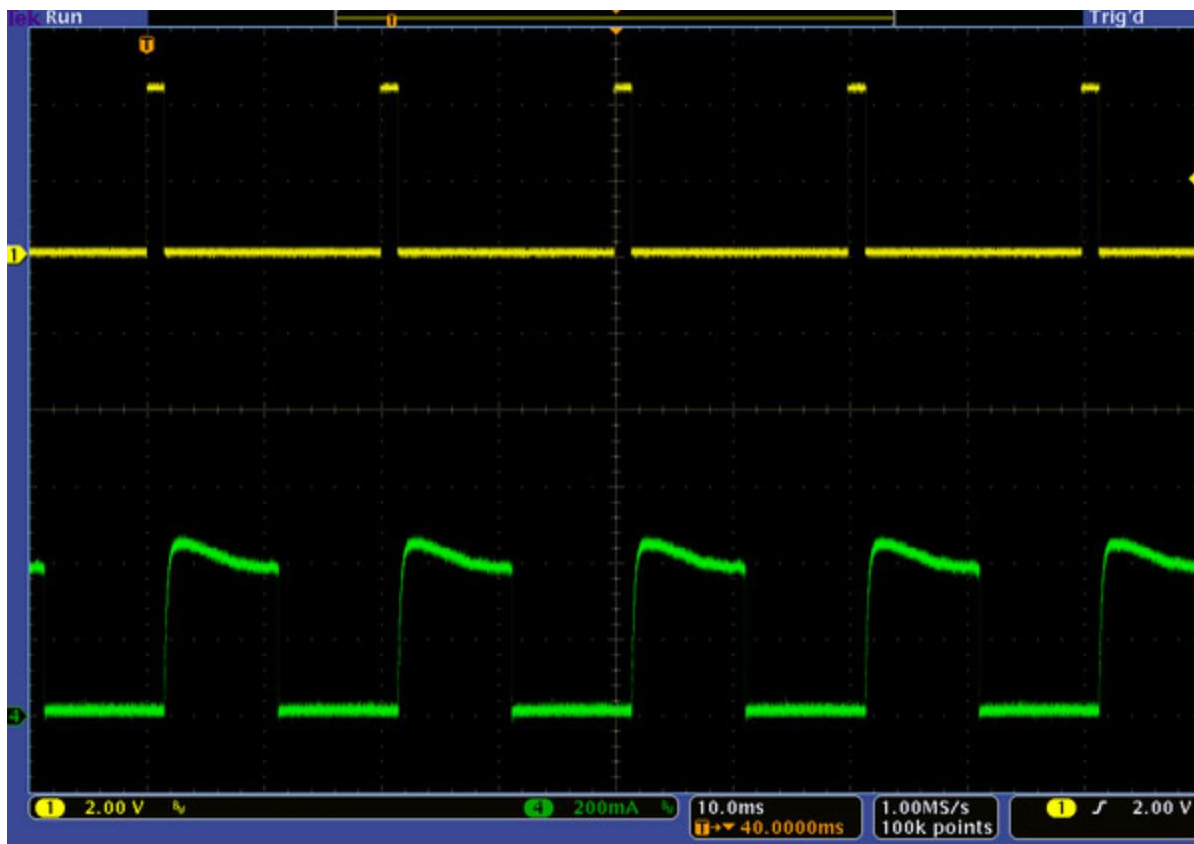
- Setup motor()
 - Set clock gating for ports B and C
 - Set TPMSRC to OSCERCLK
 - Set Duty Cycle to an appropriate value for both motors
 - Drive pins PortB3 and PortB2 with TPM0
 - Start clock with PCR PCR1 and PCR2 |= 1 << 8
- Sw1 setup()
 - Enable Port C clock
 - Enable pullups and interrupt on falling edge with PCR3 |= 0xF0703 & (0xA << 16 | (1 << 8) | (0x3))
 - Enable IRQHandler upon switch press
- Ultrasonic sensor setup()
 - Enable clock gating for Port D and A
 - Set PortA pin 13 as input
 - Set PortD pin 2 as output
 - Enable TMP1 and set to OSCERCLK
- Servo setup()
 - Enable clk gatig for TPM1 and set to OSCERCLK
 - Drive PortA pin 12 with TMP1
 - Set Duty cycle to appropriate value
 - Set prescalar and mod value
 - Start clk with TPM1->SC
- PortC IRQHandler()
 - Set servo to degree 0
 - If distance > some distance
 - Go straight and stop
 - Else
 - Turn left
 - Set servo degree 90
 - If distance > some distance
 - Turn left
 - Else
 - Turn right
- Get distance()
 - set PTD2 high
 - wait some time, set PTD2 low

- wait for echo to go high
- tStart is the time when echo goes high
- clear TPM1 TOF flag
- wait for echo to go low
- if TPM1 Overflow flag is set
- clear TPM1 TOF flag
- Set servo()
 - If degree > 180, set degree to 180
 - If degree < 0, set degree to 0
 - If degree == 0, duty cycle to 100
 - Else duty cycle is = degree * some factor

My design worked in the desired way, although I accidentally made the robot turn right instead of left most of the way through the path. My algorithms could be improved by making my turns more of a right degree turn instead of a slight turn. Despite this, my robot is still able to detect the obstacle and move in the desired direction, although it takes longer and wider turns. I conducted an experiment by placing multiple books as obstacles that the robot had to traverse.

Results:

Video: <https://photos.app.goo.gl/iA1si58Qqsw1bZzF9>



Discussions: This project was a bit frustrating because knowing how to calculate the distance using the sensor was very tedious and difficult to get right. Getting the servo to turn to 0 degrees to 90 degrees

and return to 90 degrees exactly was a big challenge. I debugged the system by incrementally changing parameters that gave the result I wanted. Also, to do all this testing and debugging in one week was extremely hard. A long due date would have helped. A lot of my other classes also had large projects due during that week.